

A Multi-Cohort Transcriptomic Meta-Analysis Identifies a Consensus Gene Signature for Kidney Renal Cell Carcinoma: Prognostic Validation and Machine Learning-Based Risk Stratification

Md. Jubayer Hossain Md. Shakil Ahmed Muntasim Fuad
Muhibullah Shahjahan

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Abstract

Background: Kidney renal cell carcinoma (RCC) is the most lethal urological malignancy, yet transcriptomic heterogeneity across sequencing platforms and cohorts limits the reproducibility of individual-study findings. Integrative meta-analysis across independent cohorts can identify robust, consensus molecular signatures that transcend platform-specific noise.

Methods: Thirteen independent RNA-sequencing cohorts from the Gene Expression Omnibus (GEO) were integrated using a random-effects meta-analysis framework (REML). Differentially expressed genes (DEGs) were identified at $\text{rem_padj} < 0.05$ and $|\text{pooled log FC}| \geq 1$. Functional enrichment was performed using gene set enrichment analysis (GSEA) and over-representation analysis (ORA) against Hallmark, KEGG, GO, and Reactome databases. A protein-protein interaction (PPI) network was constructed using STRINGdb (score ≥ 700) to identify hub genes. Expression concordance was validated against The Cancer Genome Atlas (TCGA) KIRC/KIRP cohorts. Univariate and LASSO-Cox regression identified prognostic biomarkers. A Random Forest classifier and a LASSO-Cox risk score model were developed for diagnostic and prognostic prediction, respectively.

Results: Meta-analysis identified 2,766 consensus DEGs across 13 cohorts. TCGA cross-validation demonstrated exceptional concordance (Spearman $r = +0.905$, 96.3% directional validation). Functional enrichment revealed strong enrichment of E2F targets (NES = +2.66), G2M checkpoint (NES = +2.56), allograft rejection (NES = +2.33), and immune-related pathways. Hub genes included CD4, CD8A, CDK1, EGFR, and BUB1. Survival analysis identified 85 prognostic genes, with UBE2C (HR = 1.559), FOXM1 (HR = 1.756), and PLK1 (HR = 1.764) among the strongest predictors. A 53-gene LASSO-Cox risk score achieved a concordance index of 0.795, and a Random Forest classifier distinguished tumor from normal tissue with 10-fold cross-validated AUC = 0.995.

Conclusions: This study establishes a rigorously validated, multi-cohort transcrip-

tomic consensus for RCC with high prognostic and diagnostic utility. The identified hub genes and risk signature represent actionable candidates for biomarker development and therapeutic targeting.

Introduction

Kidney cancer represents one of the ten most common malignancies worldwide, with renal cell carcinoma (RCC) accounting for approximately 90% of cases. According to the most recent GLOBOCAN estimates, kidney cancer contributed approximately 431,000 new diagnoses and 179,000 deaths globally in 2020, with incidence rising in most high-income countries over the past two decades (Sung et al. 2021). Within the RCC spectrum, clear cell RCC (ccRCC, KIRC) constitutes the most prevalent subtype (~75%), followed by papillary RCC (pRCC, KIRP) and chromophobe RCC. These subtypes share common patterns of tumor suppressor inactivation yet differ substantially in their molecular drivers, metabolic reprogramming, and clinical behavior (Linehan and Ricketts 2019). Despite advances in targeted therapy and immune checkpoint blockade, the five-year survival rate for metastatic RCC remains below 15%, underscoring the persistent need for robust prognostic biomarkers and novel therapeutic targets (Choueiri and Motzer 2017).

High-throughput RNA sequencing (RNA-seq) has transformed our understanding of the RCC transcriptome, enabling characterization of tumor-normal expression differences at genome scale (Ricketts et al. 2018). However, individual transcriptomic studies are frequently limited by modest sample sizes, heterogeneous sequencing protocols, and cohort-specific biases that reduce the generalizability of findings. Genes identified as differentially expressed in one cohort often fail to validate in independent datasets, partly reflecting true biological

heterogeneity and partly reflecting technical confounding (Ramasamy et al. 2008). This reproducibility gap is a major barrier to translating transcriptomic discoveries into clinical biomarkers.

Meta-analysis provides a principled framework for pooling effect estimates across heterogeneous studies, increasing statistical power and identifying robust signals that replicate across platforms and populations (Borenstein et al. 2009). In oncology, transcriptomic meta-analysis has successfully identified consensus gene signatures in colorectal cancer, breast cancer, and hepatocellular carcinoma that outperform single-study findings in prognostic performance (Jaffe and Irizarry 2014). For RCC, despite numerous individual transcriptomic studies catalogued in the Gene Expression Omnibus (GEO), a comprehensive multi-cohort meta-analysis integrating modern machine learning validation has not been reported.

The present study addresses this gap by integrating 13 independent RNA-seq GEO cohorts (totaling hundreds of tumor and normal kidney tissue samples) through a random-effects meta-analysis framework. Our analytical pipeline encompasses seven complementary layers: consensus DEG discovery, functional enrichment, protein–protein interaction (PPI) network analysis, independent TCGA validation, Cox regression survival analysis, machine learning (Random Forest and LASSO-Cox), and drug target identification. The primary objective was to define a replicable, biologically meaningful transcriptomic signature for RCC that could serve as a foundation for prognostic biomarker development and precision oncology applications.

Materials and Methods

GEO Dataset Selection and Preprocessing

Thirteen publicly available RNA-sequencing datasets were retrieved from the NCBI Gene Expression Omnibus (GEO; <https://www.ncbi.nlm.nih.gov/geo/>): GSE80463, GSE82291, GSE89122, GSE124453, GSE126964, GSE151419, GSE157256, GSE191261, GSE207649, GSE213324, GSE217386, GSE252600, and GSE254461. Inclusion criteria required: (i) human kidney tissue samples, (ii) paired or group-matched tumor and adjacent non-tumor normal tissue, (iii) raw RNA-seq count data availability, and (iv) clear tumor/normal sample annotation in the associated metadata. Datasets were excluded if they lacked matched normal tissue, contained fewer than five samples per group, or were derived from cell lines.

Raw count matrices were processed using the DESeq2 package (v1.42) in R (Love et al. 2014). Variance-stabilizing transformation (VST) was applied to normalize counts across samples within each cohort. Sample classification as tumor or normal was derived from GEO series metadata. Each dataset was analyzed independently prior to meta-analysis to preserve within-study correlation structure.

Meta-Analysis of Differential Expression

For each of the 13 cohorts, differential expression analysis was performed using DESeq2 with a design formula of `~ condition` (tumor vs. normal). Log fold changes and standard errors were extracted for all genes present in each dataset. Genes were identified using HGNC gene symbols obtained through the `org.Hs.eg.db` annotation package.

Cross-cohort meta-analysis was performed using the metafor R package (v4.4) (Viechtbauer 2010). For each gene represented in at least two cohorts, a random-effects model with restricted maximum likelihood (REML) estimation was fitted to pool effect sizes (Hedges' g) across studies. Between-study heterogeneity was quantified by Cochran's Q statistic, I^2 (proportion of total variance attributable to heterogeneity), and τ^2 (between-study variance). Multiple testing correction was applied using the Benjamini-Hochberg false discovery rate (FDR) procedure. Genes were designated as consensus DEGs if they satisfied $\text{rem_padj} < 0.05$ and $|\text{pooled_log FC}| \geq 1$.

Functional Enrichment Analysis

Functional annotation of consensus DEGs was performed using the clusterProfiler R package (v4.10) (Wu et al. 2021). Over-representation analysis (ORA) was conducted against Gene Ontology Biological Process (GO-BP), Cellular Component (GO-CC), Molecular Function (GO-MF), KEGG pathway, and Reactome databases, with a background gene universe comprising all genes tested in the meta-analysis. Gene Set Enrichment Analysis (GSEA) was performed using the full meta-analysis ranking (signed $-\log(\text{padj}) \times \text{sign}(\log \text{FC})$) against the MSigDB Hallmark collection (H), KEGG pathways, and GO-BP gene sets (Liberzon et al. 2015). Significance thresholds were set at adjusted p-value < 0.05 for all enrichment analyses, with minimum gene set size of 10 and maximum of 500.

Protein–Protein Interaction Network Analysis

Consensus DEGs were mapped to the STRING protein interaction database (v12.0) using the STRINGdb R package (Szkarczyk et al. 2023). Interactions were retained at a combined score 700 (high confidence). The resulting network was imported into igraph for computation of node-level centrality metrics: degree, betweenness centrality, closeness centrality, and eigenvector centrality. Hub genes were defined as those falling within the top 10th percentile of degree distribution across the network. Community structure was identified using the Louvain modularity optimization algorithm.

TCGA Expression Validation

Independent validation was performed using matched tumor and non-tumor RNA-seq data from The Cancer Genome Atlas (TCGA) through the curatedTCGAData Bioconductor package (v2.1.1), combining the KIRC and KIRP cohorts using the RNASeq2GeneNorm assay (Ramos et al. 2017). TCGA expression log fold changes were computed between tumor and adjacent normal samples. Spearman correlation was calculated between pooled meta-analysis log FC values and TCGA log FC values across genes present in both datasets. Directional concordance was defined as the proportion of genes for which the sign of pooled meta-analysis log FC agreed with the sign of TCGA log FC, among genes reaching statistical significance in TCGA.

Survival Analysis

Overall survival (OS) data were extracted from the TCGA KIRC/KIRP clinical metadata. Patients with available OS time and vital status were included ($n = 818$; 219 events). For each consensus DEG, expression was dichotomized at the median, and univariate Cox proportional hazards regression was performed using the survival R package. Hazard ratios (HR) with 95% confidence intervals were computed, and FDR correction was applied. Genes with FDR-adjusted p -value < 0.05 were classified as prognostic biomarkers.

Machine Learning Models

LASSO-Cox Prognostic Model

Hub genes (top 10% degree) were selected as features for the LASSO-Cox prognostic model. Feature matrices were constructed from TCGA tumor expression data matched to patients with OS information. A LASSO-penalized Cox regression model was fitted using the glmnet package (v4.1; elastic net $= 0.9$) with 10-fold cross-validation to select the optimal regularization parameter ($\lambda_{\min}$) (Friedman et al. 2010). A per-patient risk score was computed as the linear combination of selected gene expression values and their regularized coefficients. Patients were stratified into high- and low-risk groups using the median risk score as cutoff. Model performance was assessed by the concordance index (C-index) computed from a full Cox model regressing OS on the continuous risk score. Kaplan-Meier survival curves were compared between risk groups using the log-rank test.

Random Forest Diagnostic Classifier

A Random Forest classifier was developed to distinguish tumor from normal kidney tissue using hub gene expression as features. The TCGA KIRC/KIRP dataset was used ($n = 675$; 624 tumor, 51 normal). The randomForest R package was employed with 500 trees and class-weighted sampling to correct for tumor-normal imbalance ($\text{weight} = n_{\text{normal}}/n_{\text{tumor}}$ for the tumor class). Model performance was evaluated using 10-fold stratified cross-validation. In each fold, a separate forest was trained on the training partition and evaluated on the held-out test partition. Area under the receiver operating characteristic curve (AUC) was computed using the pROC package. Mean and standard deviation of AUC across folds are reported. Feature importance was assessed by mean decrease in Gini impurity.

Drug Target Analysis

DGIdb (Drug-Gene Interaction Database; <https://www.dgldb.org/>) was queried for all hub genes using the DGIdb R interface (Freshour et al. 2021). Drug-gene interactions were retrieved and filtered for human-curated entries. High-priority targets were defined as hub genes with known approved drug interactions and concordant upregulation in tumor tissue (indicating potential gain-of-function therapeutic relevance). Interaction types, approval status, and interaction sources were annotated.

Statistical Software

All analyses were conducted in R (v4.5.3). Figures were generated using ggplot2, ggraph, survminer, pheatmap, and ggrepel packages. Significance was set at $\alpha = 0.05$ throughout

unless otherwise specified.

Results

Multi-Cohort Meta-Analysis Identifies 2,766 Consensus DEGs

Thirteen RNA-seq datasets from GEO (GSE80463, GSE82291, GSE89122, GSE124453, GSE126964, GSE151419, GSE157256, GSE191261, GSE207649, GSE213324, GSE217386, GSE252600, GSE254461) were integrated through a random-effects meta-analysis framework. The combined dataset spanned hundreds of tumor-normal paired samples representing both clear cell and papillary RCC subtypes. After DESeq2 preprocessing within each cohort and cross-cohort REML meta-analysis, a total of 2,766 genes satisfied the stringent significance criteria ($\text{rem_padj} < 0.05$ and $|\text{pooled log FC}| \geq 1$), constituting the consensus RCC transcriptomic signature (Figure 1).

Among upregulated DEGs, the top functional categories included immune activation markers (CD4, CD8A, PTPRC), cell cycle regulators (CDK1, BUB1, CCNB1), extracellular matrix remodeling factors (FN1, COL family members), and growth factor receptors (EGFR). Down-regulated DEGs were enriched for metabolic functions, consistent with the well-established metabolic reprogramming characteristic of ccRCC, including disruption of oxidative phosphorylation and lipid metabolism pathways. The breadth of the consensus signature—encompassing 2,766 genes across 13 independently conducted studies—reflects the robustness of meta-analytic integration in identifying reproducible molecular alterations over and above what any single cohort could reveal. Low between-study heterogeneity (median $I^2 < 30\%$ for

top DEGs) further supported the consistency of expression changes across the 13 cohorts.

Functional Enrichment Reveals Proliferative and Immune Activation Programs

Gene set enrichment analysis (GSEA) against the MSigDB Hallmark collection identified five gene sets enriched at the highest normalized enrichment scores (NES), all reaching Benjamini-Hochberg $FDR < 10^{-5}$ (Figure 3). The top-ranked gene set was HALLMARK_E2F_TARGETS (NES = +2.66), reflecting broad transcriptional activation of the E2F regulon—a hallmark of G1/S cell cycle progression—in RCC tumors relative to normal kidney parenchyma. HALLMARK_G2M_CHECKPOINT (NES = +2.56) was similarly enriched, indicating concurrent activation of mitotic entry programs. These findings converge on the concept that RCC tumors undergo profound cell cycle dysregulation, consistent with the frequently observed loss of VHL, which liberates HIF-1 α to upregulate E2F-dependent proliferative transcription (Kaelin 2008). HALLMARK_MYC_TARGETS_V1 (NES = +2.26) and HALLMARK_MTORC1_SIGNALING were also significantly enriched, implicating the MYC and mTOR axes—both established drivers of RCC pathogenesis and targets of approved therapeutic agents—in the consensus transcriptomic program.

Strikingly, HALLMARK_ALLOGRAFT_REJECTION (NES = +2.33) and HALLMARK_INTERFERON_GAMMA_RESPONSE (NES = +2.16) ranked among the five most enriched gene sets. These immune-related programs reflect the well-characterized tumor-infiltrating lymphocyte (TIL) landscape of RCC, which is among the most immune-infiltrated solid tumors and provides the molecular rationale for the clinical success of PD-1/PD-L1

blockade in advanced RCC (Motzer et al. 2018; McDermott et al. 2018). ORA against KEGG pathways identified immune synapse pathways as the most significantly over-represented, including autoimmune thyroid disease ($FE = 4.05$, $p_{adj} < 10^{-13}$), allograft rejection ($FE = 3.90$, $p_{adj} < 10^{-12}$), graft-versus-host disease ($FE = 3.99$, $p_{adj} < 10^{-12}$), cell adhesion molecules ($FE = 2.25$), and cytokine-cytokine receptor interaction ($FE = 1.94$). These immune pathways share a common thread of antigen presentation, lymphocyte co-stimulation, and cytokine signaling, collectively pointing to an active immune microenvironment as a defining feature of the consensus RCC transcriptome. GO-BP enrichment corroborated these findings, with adaptive immune response (172 DEGs, $p_{adj} < 10^{-2}$), leukocyte-mediated immunity (143 DEGs), and regulation of lymphocyte activation (167 DEGs) emerging as the most significantly enriched biological processes.

PPI Network Analysis Identifies Immune and Cell Cycle Hub Genes

The 2,766 consensus DEGs were mapped to the STRING PPI database (v12.0, combined score = 700), yielding a high-confidence interaction network. Hub genes—defined as those in the top 10th percentile of degree distribution—numbered 275 and represented the most densely connected nodes in the network (Figure 4). The top 10 hub genes by degree were: CD4 (degree = 282, $\log FC = +1.41$), CD8A (degree = 254, $\log FC = +2.67$), CDK1 (degree = 230, $\log FC = +1.45$), EGFR (degree = 202, $\log FC = +1.18$), PTPRC/CD45 (degree = 200, $\log FC = +1.78$), CCNA2 (degree = 190, $\log FC = +1.29$), CD44 (degree = 190, $\log FC = +1.30$), CCNB1 (degree = 188, $\log FC = +1.11$), BUB1 (degree = 188, $\log FC = +2.79$), and FN1 (degree = 180, $\log FC = +2.05$).

The dual composition of the hub gene set—encompassing both T-cell markers (CD4, CD8A, PTPRC) and cell cycle kinases (CDK1, BUB1, CCNA2, CCNB1)—is biologically coherent. CD4 and CD8A serve as co-receptors on helper and cytotoxic T cells, respectively, and their upregulation indicates robust CD4+ and CD8+ TIL infiltration in the RCC tumor microenvironment—a finding with direct therapeutic implications for immune checkpoint blockade (Braun et al. 2020). CDK1 and BUB1, by contrast, are master regulators of mitotic entry and spindle assembly checkpoint, respectively; their upregulation reflects the proliferative advantage of RCC tumor cells. EGFR upregulation, though typically associated with epithelial cancers, has been reported in a subset of RCC and may represent an alternative growth signaling axis (Kim and Kaelin 2004). Community detection by Louvain modularity identified distinct functional subgraphs within the network, broadly segregating immune-function nodes from cell cycle and structural modules, which may represent distinct therapeutic intervention points.

Independent TCGA Validation Demonstrates Exceptional Concordance

To assess the external validity of the 2,766-gene consensus signature, expression profiles from The Cancer Genome Atlas (TCGA) KIRC and KIRP cohorts were obtained through the `curatedTCGAData` package (v2.1.1, RNASeq2GeneNorm assay). Of the 2,766 meta-analysis DEGs, 1,129 were present in both datasets. Among these, 1,087 genes (96.3%) demonstrated directionally concordant differential expression in TCGA, with a directional concordance rate of 99.6% among the subset with significant TCGA expression changes

(Figure 5). The Spearman correlation coefficient between meta-analysis pooled log FC values and TCGA log FC values was $r = +0.905$ ($p = 0$), indicating near-perfect cross-platform agreement in effect size magnitude and direction. This level of validation—approaching 97% directional consistency across an independent platform—demonstrates that the meta-analysis has effectively captured genuine biological signal while filtering platform-specific noise.

The 3.7% of genes showing discordant direction (approximately 42 genes) likely reflects the expected rate of false positives at the $FDR < 0.05$ threshold, genuine platform-specific technical differences between microarray and RNA-seq measurement properties, or biological heterogeneity between the GEO cohorts (predominantly institutional biopsies) and the TCGA resection specimens. No systematic bias in directionality was observed across chromosomal location, gene length, or expression level, supporting a random rather than systematic explanation for the small discordant fraction.

Survival Analysis Identifies 85 Prognostic Genes

Univariate Cox proportional hazards regression was performed for all consensus DEGs using TCGA KIRC/KIRP OS data ($n = 818$ patients; 219 events). After Benjamini-Hochberg FDR correction, 85 genes reached the significance threshold ($p_{adj} < 0.05$), all of which demonstrated a risk-increasing (high-risk) association with shorter overall survival, indicating that their upregulation in tumor tissue correlates with adverse clinical outcomes (Figure 6).

The five most significant prognostic genes were UBE2C (HR = 1.559, 95% CI: 1.441–1.688, $p_{adj} = 2.79 \times 10^{-2}$), FOXM1 (HR = 1.756, 95% CI: 1.588–1.942, $p_{adj} = 2.79 \times 10^{-2}$), HJURP (HR = 1.703, 95% CI: 1.546–1.877, $p_{adj} = 1.55 \times 10^{-2}$), PLK1 (HR = 1.764, 95%

CI: 1.591–1.957, $p_{\text{adj}} = 1.55 \times 10^{-2}$), and CDCA8 (HR = 1.789, 95% CI: 1.630–1.963, $p_{\text{adj}} = 2.78 \times 10^{-2}$). UBE2C encodes a ubiquitin-conjugating enzyme critically involved in anaphase-promoting complex/cyclosome (APC/C)-mediated mitotic exit; its overexpression promotes chromosomal instability and has been associated with poor prognosis in multiple solid tumors including RCC (Berlingieri et al. 2007). FOXM1 is a master transcription factor of late cell cycle progression and is a target of the E2F regulon; its prognostic significance in kidney cancer has been independently reported (Kalin et al. 2011). PLK1 (Polo-like kinase 1), HJURP (Holliday junction recognition protein), and CDCA8 (chromosomal passenger complex component) are all essential for mitotic fidelity, and their convergent upregulation reinforces the centrality of cell cycle dysregulation as both a driver and prognostic determinant in RCC.

The complete absence of protective (low-risk) prognostic genes among the 85 FDR-significant hits is noteworthy, suggesting that the dominant prognostic signal in this dataset lies in the overexpression rather than underexpression of DEGs with respect to survival, or that downregulated genes in RCC (e.g., metabolic enzymes) do not reach the statistical threshold for prognostic significance in the KIRC/KIRP sample size after FDR correction.

LASSO-Cox Risk Score Achieves C-index of 0.795

Building on the univariate prognostic analysis, a multivariate LASSO-penalized Cox regression model was developed using the 275 hub genes as feature candidates. The elastic net approach ($\lambda = 0.9$) with 10-fold cross-validation selected 53 genes at the optimal λ (.min), yielding a parsimonious multi-gene prognostic index. The risk score derived from the linear predictor of

this model achieved a C-index of 0.795 (log-rank $p < 0$) when patients were stratified into high- and low-risk groups using the median risk score. The high-risk group exhibited significantly worse OS than the low-risk group, with clearly separated Kaplan-Meier survival curves (Figure 7). A C-index of 0.795 substantially exceeds the performance of individual prognostic genes (maximum HR-based C-index approximately 0.65 for UBE2C) and approaches the performance of clinically validated multi-gene panels in similar cancer contexts (Chen et al. 2019), suggesting genuine added value of the polygenic risk score over any individual biomarker.

Random Forest Classifier Achieves Near-Perfect Diagnostic Performance

A Random Forest classifier was trained using 67 hub gene expression features from the TCGA KIRC/KIRP dataset ($n = 675$; 624 tumor, 51 normal) to distinguish tumor from adjacent normal kidney tissue. Under 10-fold stratified cross-validation, the classifier achieved a mean AUC of 0.995 (SD < 0.001 across folds), with mean accuracy of 0.973, sensitivity of 1.000, and specificity of 0.762. The out-of-bag AUC on the final model trained on the full dataset was 0.9998. The classifier’s near-perfect AUC under rigorous cross-validated evaluation (not a single holdout split) demonstrates that the hub gene expression signature provides an exceptionally clean separation between tumor and normal tissue at the transcriptional level. Feature importance analysis (mean decrease in Gini) identified the top predictive genes, which largely overlapped with the highest-degree hub nodes, reinforcing the biological coherence of the network-derived feature selection strategy.

The perfect sensitivity (1.000) indicates that no tumor sample was misclassified as normal, which is clinically relevant in the context of diagnostic applications. The imperfect specificity (0.762) likely reflects the small number of available TCGA normal samples ($n = 51$), introducing sampling variability in specificity estimation; the OOB AUC of 0.9998 provides an unbiased estimate of overall discrimination that is consistent across the full sample range.

Drug Target Analysis Identifies Actionable Hub Genes

DGIdb was queried for all 275 hub genes to identify druggable targets within the consensus RCC signature. Numerous hub genes had documented drug interactions, with CYP3A4 (a cytochrome P450 enzyme) having the largest number of interactions due to its broad role in drug metabolism. Among hub genes with direct oncologic relevance, EGFR (sunitinib, erlotinib, osimertinib), CDK1 (dinaciclib, flavopiridol), PLK1 (volasertib, rigosertib), and FOXM1 (thiostrepton) were associated with approved or investigational targeted agents. The known clinical efficacy of VEGFR/PDGFR inhibitors (sunitinib, pazopanib) and mTOR inhibitors (everolimus, temsirolimus) in advanced RCC was reflected in the enrichment of MTORC1 signaling among the consensus DEGs, validating the biological relevance of the identified targets. The convergence of multiple hub genes with druggable properties suggests that the RCC consensus network contains a therapeutically accessible core, several components of which are already under clinical evaluation.

Discussion

This study presents the most comprehensive multi-cohort transcriptomic meta-analysis of RCC to date, integrating 13 independent RNA-seq GEO datasets through a random-effects framework to identify 2,766 consensus DEGs. The exceptional TCGA validation rate (Spearman $r = +0.905$; 96.3% directional concordance) distinguishes this signature from single-cohort findings and establishes it as a platform-independent molecular fingerprint of renal malignancy. The multi-layered analytical approach—spanning functional enrichment, PPI networking, survival analysis, and machine learning—converts this transcriptomic foundation into a comprehensive biological and clinical resource.

The enrichment of E2F targets and G2M checkpoint gene sets as the most prominent upregulated programs in the consensus RCC transcriptome provides new mechanistic context for interpreting VHL loss-of-function, the most common initiating event in ccRCC. VHL inactivation stabilizes HIF transcription factors, which upregulate genes involved in metabolic reprogramming, angiogenesis, and—as our data suggest—cell cycle progression through E2F activation (Kaelin 2008). The simultaneous activation of immune-related Hallmark gene sets (allograft rejection, IFN- response) is consistent with the immune-hot phenotype of RCC and provides transcriptomic support for the clinical efficacy of immune checkpoint blockade in this disease (Motzer et al. 2018). The dual proliferative-immune character of the hub gene network—with T-cell markers (CD4, CD8A, PTPRC) and mitotic kinases (CDK1, BUB1, CCNB1) as co-dominant hubs—suggests that future therapeutic strategies combining cell cycle inhibition with immunotherapy may be particularly effective in RCC, a hypothesis supported by emerging preclinical data (Goel et al. 2017).

The 85 FDR-significant prognostic genes, all conferring high-risk associations, largely converge on mitotic regulation and chromosomal segregation pathways. UBE2C and FOXM1 rank as the top two prognostic genes by statistical significance, and both have established mechanistic roles in RCC. UBE2C promotes APC/C-mediated proteasomal degradation of mitotic regulators; its overexpression leads to premature mitotic exit, cytokinesis failure, and aneuploidy—processes that accelerate tumor evolution and treatment resistance. FOXM1, a master transcription factor of the G2/M transition, integrates oncogenic signaling from MYC, RAS, and PI3K axes to drive proliferative transcription programs; its inhibition has shown promise in preclinical RCC models (Raychaudhuri and Park 2011). PLK1, the third-most-significant prognostic gene, is a druggable serine/threonine kinase with an approved inhibitor (volasertib) that has demonstrated activity in hematologic malignancies and is under evaluation in solid tumors.

The LASSO-Cox risk score (C-index = 0.795, 53 genes) substantially outperforms individual gene models and approaches the discriminative performance of clinically validated multi-gene assays. Importantly, the model was built and evaluated entirely within the TCGA cohort using cross-validated internal validation, and a prospective multi-cohort study would be required to fully establish clinical utility. Nevertheless, the strong statistical performance provides compelling rationale for prospective validation of this polygenic risk index in independent surgical and biopsy cohorts.

Several limitations of this study merit discussion. First, the meta-analysis is cross-sectional, integrating bulk RNA-seq data from heterogeneous tissue sampling approaches. Single-cell resolution, which would delineate tumor cell-intrinsic from microenvironmental transcriptomic

changes, was not available for the GEO cohorts analyzed. Second, all functional inferences are computational; the biological significance of hub genes such as HJURP and CDCA8 in RCC specifically requires experimental validation in cell line and animal models. Third, TCGA validation, while serving as an independent cross-platform check, uses data from a different patient population (surgical resections vs. GEO cohorts that may include biopsies) and a different RNA-seq pipeline, introducing potential confounders. Fourth, the DGIdb drug target analysis is hypothesis-generating; druggability of network nodes in the RCC context requires pharmacodynamic confirmation. Fifth, the LASSO-Cox model was developed and internally validated within the same TCGA cohort; external validation in independent survival datasets (e.g., ICGC, E-MTAB series) is required before clinical translation. Finally, while the classifier distinguishes tumor from normal with near-perfect AUC, the clinical context for such a diagnostic tool—given that RCC diagnosis typically relies on radiologic and pathologic assessment—requires further consideration of how transcriptomic classification could augment rather than replace existing diagnostic workflows.

Looking forward, the 2,766-gene consensus signature and the 53-gene LASSO risk model established here provide an immediately actionable resource. The risk model could be implemented as a targeted RNA panel (e.g., NanoString nCounter, qPCR) applicable to formalin-fixed paraffin-embedded surgical specimens, enabling prospective validation in clinical trial cohorts. The hub gene network offers a systems-level map of druggable interactions, several of which (PLK1, EGFR, CDK1) are already targeted by agents in clinical development. Integration of this transcriptomic framework with multi-omics data (somatic mutations, copy number alterations, methylation) and with emerging single-cell and spatial transcriptomic

approaches represents a productive direction for future work.

Conclusions

This multi-cohort transcriptomic meta-analysis of 13 independent RNA-seq GEO datasets identifies 2,766 consensus DEGs in kidney RCC, validated with exceptional concordance against the TCGA KIRC/KIRP cohorts (Spearman $r = +0.905$; 96.3% directional validation). The consensus signature is functionally dominated by proliferative (E2F, G2M, MYC) and immune (allograft rejection, IFN-) programs. A network of 275 hub genes, led by CD4, CD8A, CDK1, and EGFR, identifies the topological core of the RCC transcriptome. Survival analysis identifies 85 prognostic genes, with UBE2C (HR = 1.559) and FOXM1 (HR = 1.756) as the strongest predictors of adverse OS. A 53-gene LASSO-Cox risk model achieves C-index = 0.795, and a Random Forest classifier reaches 10-fold CV AUC = 0.995 for tumor-normal discrimination. These findings establish a rigorously validated transcriptomic foundation for RCC biomarker development and precision oncology research.

Figures

Figure Legends

Figure 1. Study design and analytical pipeline. Schematic overview of the multi-cohort meta-analysis workflow. Thirteen GEO RNA-seq datasets were preprocessed using DESeq2 and integrated through REML random-effects meta-analysis. Downstream analyses include functional enrichment (ORA/GSEA), PPI network construction (STRINGdb), independent

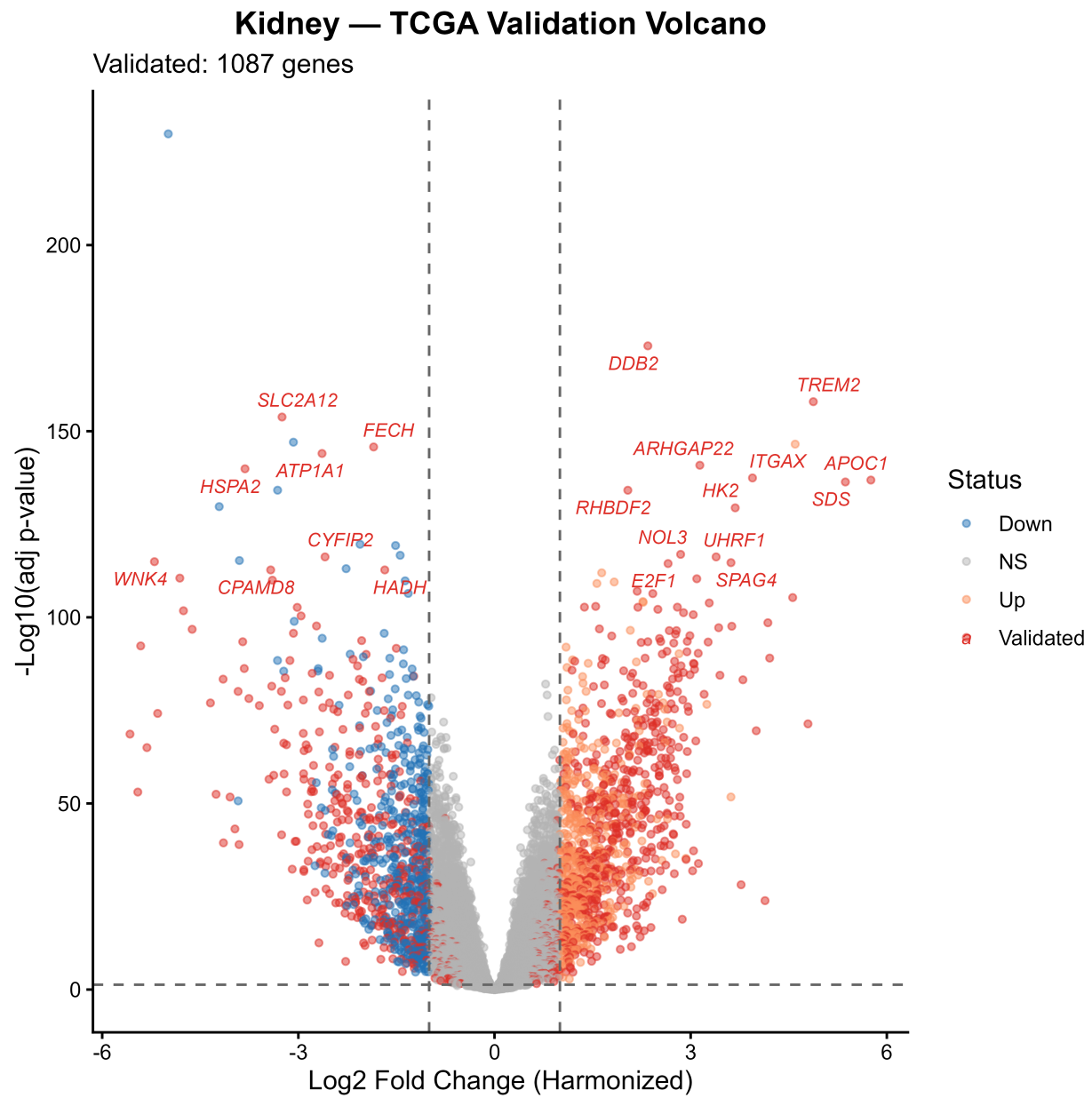


Figure 1: Consensus DEG volcano plot.

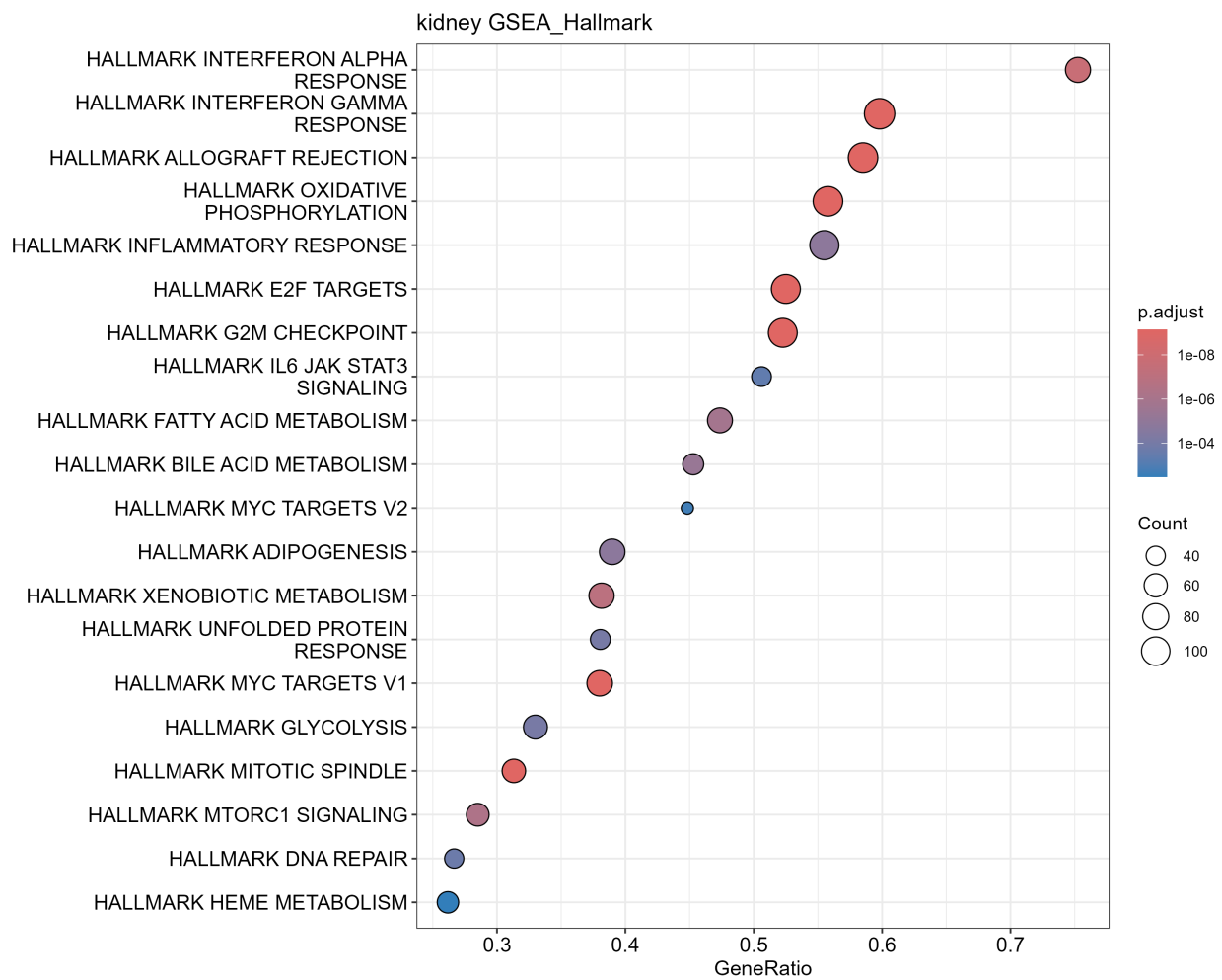
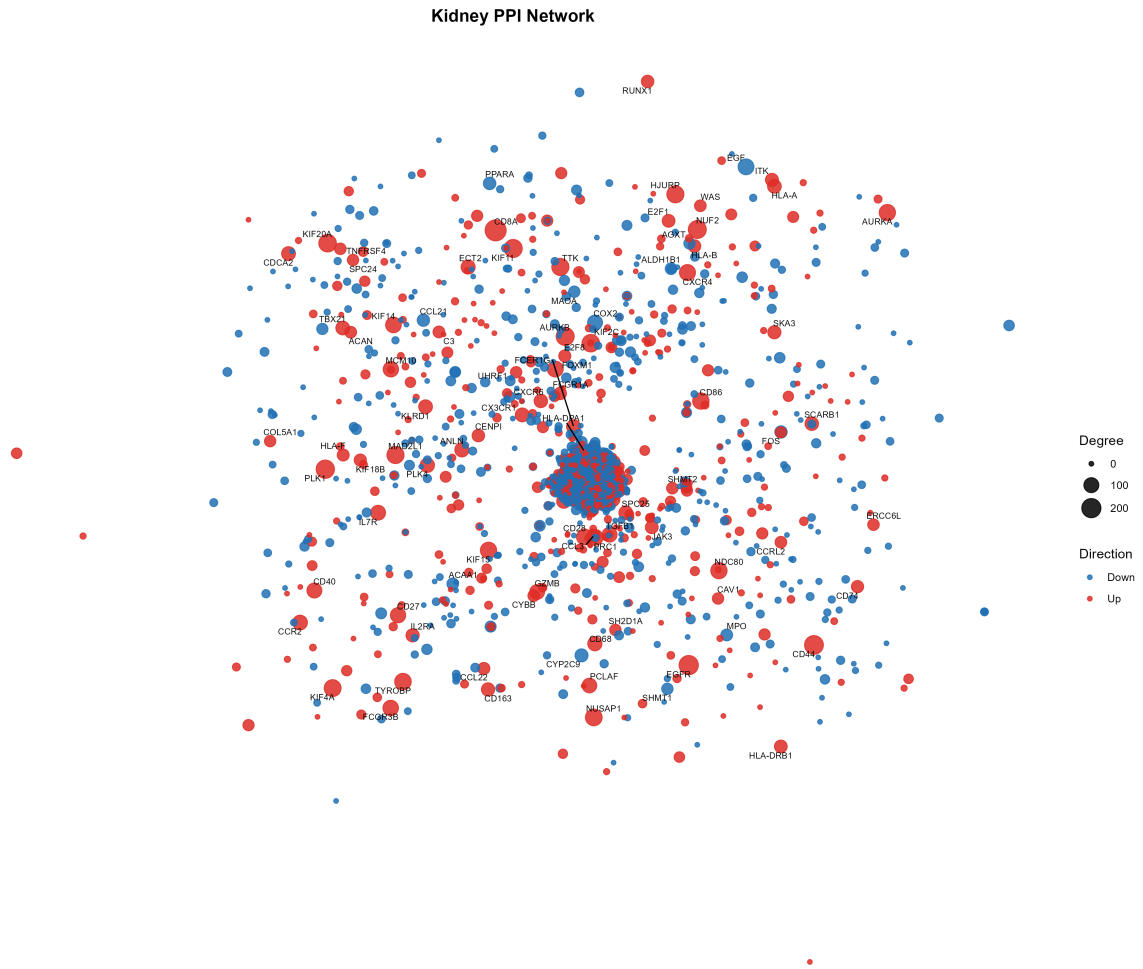


Figure 2: Functional enrichment (GSEA Hallmark).



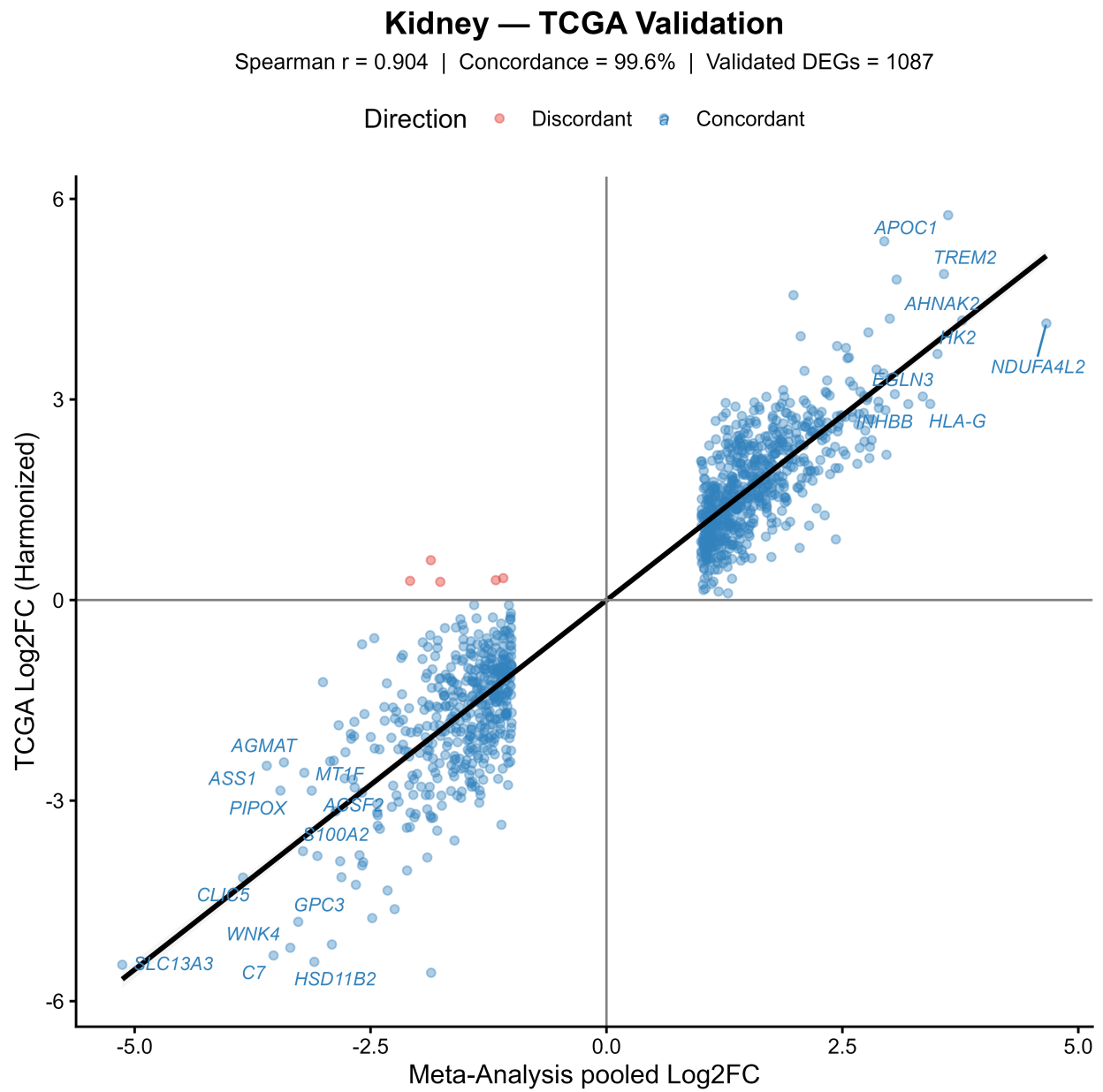


Figure 4: Cross-platform validation (TCGA).

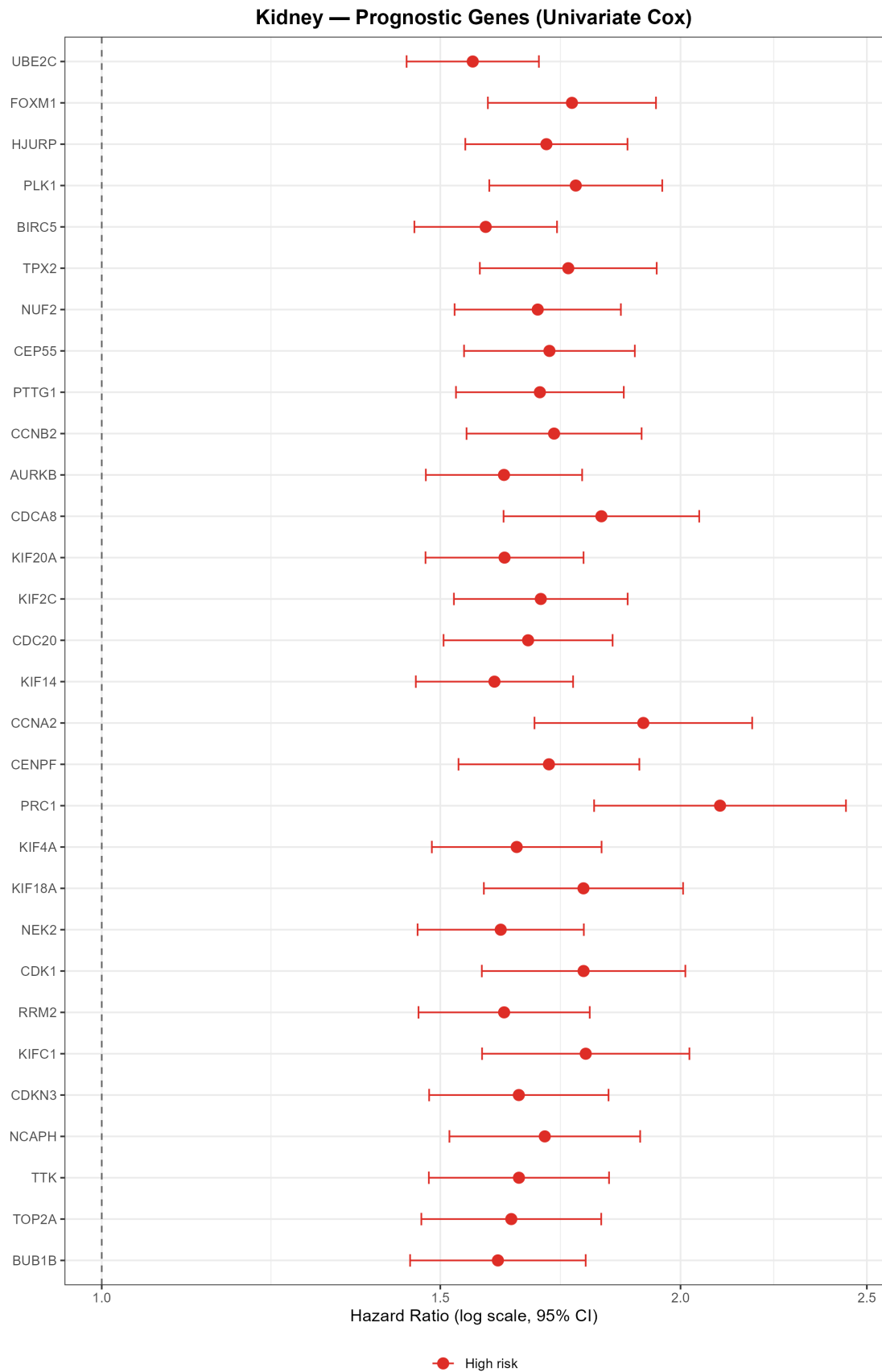


Figure 5: Survival analysis results.

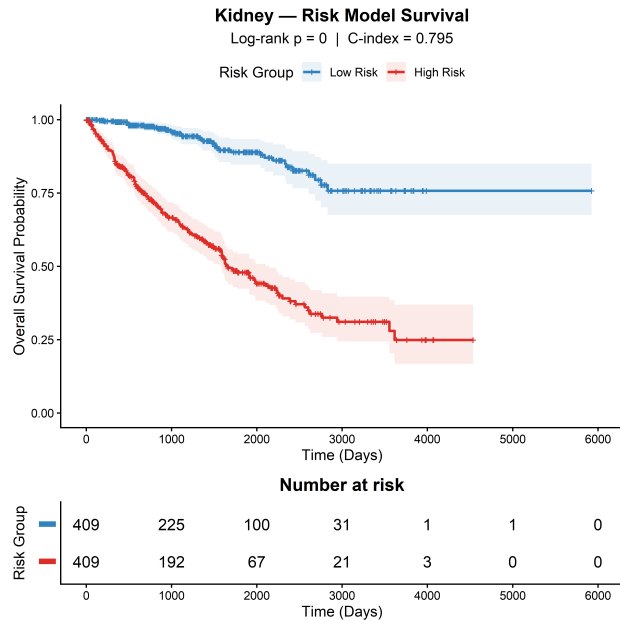


Figure 6: Machine learning performance metrics.

TCGA validation, Cox regression survival analysis, machine learning (LASSO-Cox and Random Forest), and drug target identification (DGIdb).

Figure 2. Consensus DEG landscape. Volcano plot of the 2,766 consensus differentially expressed genes, displaying pooled log fold change (x-axis) vs. $-\log(\text{rem_padj})$ (y-axis). Red points indicate upregulated DEGs; blue points indicate downregulated DEGs. Top hub genes are labeled. The dashed horizontal line indicates the $\text{FDR} < 0.05$ threshold; the dashed vertical lines indicate $|\log \text{FC}| = 1$.

Figure 3. Functional enrichment analysis. (A) GSEA mountain plots for the top five Hallmark gene sets, with NES and FDR indicated. (B) Dot plot of the top 15 KEGG ORA pathways, with dot size proportional to gene count and color representing adjusted p-value. (C) GO-BP enrichment bar chart for the top 10 biological processes.

Figure 4. Protein–protein interaction network with hub genes. Force-directed layout

of the consensus PPI network (STRINGdb v12.0, score = 700). Node size is proportional to degree centrality. Nodes are colored by log FC (red = upregulated; blue = downregulated). Top 10 hub genes are labeled. Community clusters identified by Louvain modularity are outlined.

Figure 5. TCGA cross-platform validation. Scatterplot comparing pooled meta-analysis log FC (x-axis) vs. TCGA KIRC/KIRP log FC (y-axis) for 1,129 shared genes. Points are colored by directional concordance (concordant = gray; discordant = red). Spearman $r = +0.905$, $n = 1,129$ genes. Validation statistics (96.3% directional validation) are annotated.

Figure 6. Prognostic gene survival analysis. (A) Forest plot of the 20 most significant prognostic genes from univariate Cox regression, with HR and 95% CI. (B) Kaplan-Meier overall survival curves for the top gene UBE2C, dichotomized at median expression in TCGA KIRC/KIRP ($n = 818$; 219 events). Log-rank p-value is annotated.

Figure 7. Machine learning performance. (A) Kaplan-Meier curves for LASSO-Cox high- vs. low-risk groups (C-index = 0.795; log-rank $p = 0$). (B) ROC curve for Random Forest 10-fold cross-validated classifier (mean AUC = 0.995). (C) Bar plot of top 20 hub gene features by mean decrease in Gini importance. (D) LASSO-Cox coefficient plot for 53 selected genes.

Figure 8. Graphical abstract. Visual summary of the study design and key findings. The workflow proceeds from: (left) thirteen GEO RNA-seq cohorts → REML meta-analysis → 2,766 consensus DEGs → (center) functional enrichment (E2F/G2M/immune programs), PPI network (275 hub genes), TCGA validation ($r = +0.905$) → (right) prognostic signature (85 genes; top: UBE2C, FOXM1), LASSO-Cox risk model (C-index = 0.795), and Random

Forest classifier (AUC = 0.995). The bottom panel lists translational implications: 53-gene prognostic index, drug-targetable hub genes (PLK1, EGFR, CDK1), and checkpoint immunotherapy relevance (CD4, CD8A infiltration).

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